

Linear Algebra – MAT 2610

Section 2.1 (Matrix Operations)

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Matrix Addition

In order to add two matrices, those matrices must be the same size

The sum of two $n \times m$ matrices is also an $n \times m$ matrix

We add two matrices by adding corresponding entries

Subtraction works in the same way

$$\begin{bmatrix} 3 & 1 \\ 4 & -2 \end{bmatrix} + \begin{bmatrix} 1 & -3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 3+1 & 1-3 \\ 4-1 & -2-2 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 3 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & a & 4 \\ -1 & 2 & b \end{bmatrix} - \begin{bmatrix} c & d & 6 \\ -2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 2-c & a-d & 4-6 \\ -1+2 & 2-3 & b-4 \end{bmatrix}$$
$$= \begin{bmatrix} 2-c & a-d & -2 \\ 1 & -1 & b-4 \end{bmatrix}$$



A **scalar** is a real number (or a variable representing a real number)

A scalar multiple of a matrix A refers to the matrix obtained by multiplying every entry in A by the same constant

If c is a scalar and A is $m \times n$, then the scalar multiple cA is also $m \times n$

$$2 \begin{bmatrix} 3 & 1 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 8 & -4 \end{bmatrix}$$

$$-3 \begin{bmatrix} 4 & a & -2 \\ 5 & -1 & b - 2c \end{bmatrix} = \begin{bmatrix} -12 & -3a & 6 \\ -15 & 3 & -3b + 6c \end{bmatrix}$$



Theorem 1

Let A, B, C be $m \times n$ matrices, and let s and t be any scalars. Then

a) $A + B = B + A$ — (commutative law)

b) $(A + B) + C = A + (B + C)$ — (associative law)

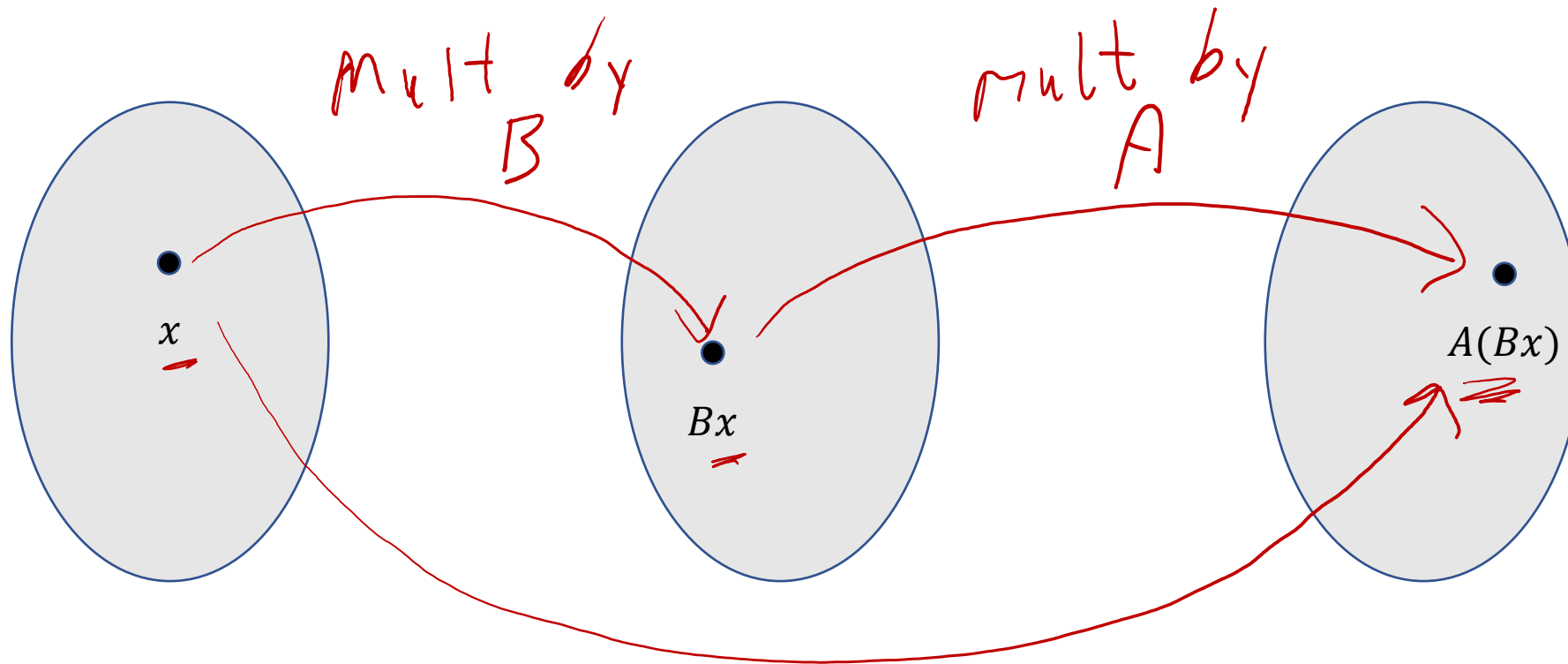
c) $A + 0 = A$ —

d) $s(A + B) = sA + sB$ —

e) $(s + t)A = sA + tA$ —

f) $(st)A = s(tA)$ —





$$\begin{aligned} Cx &= A(Bx) \\ &= (AB)x \end{aligned}$$

Composition



Definition: The product AB of the $m \times n$ matrix A and the $n \times p$ matrix B is defined as

$$AB = [Ab_1 \quad Ab_2 \quad \dots \quad Ab_p]$$

where b_i is the i^{th} column of B

Example:

$$\begin{aligned}
 & \underbrace{[2 \quad 3 \quad -2]}_{1 \times 3} \underbrace{\begin{bmatrix} r & -2 \\ s & t \\ 2 & -3 \end{bmatrix}}_{3 \times 2} = \begin{bmatrix} [2 \quad 3 \quad -2] \begin{bmatrix} r \\ s \\ 2 \end{bmatrix} \\ [2 \quad 3 \quad -2] \begin{bmatrix} -2 \\ t \\ -3 \end{bmatrix} \end{bmatrix} \\
 & = \begin{bmatrix} 2r + 3s - 4 & -4 + 3t + 6 \\ 2r + 3s - 4 & 2 + 3t \end{bmatrix}
 \end{aligned}$$



Note: The product AB of the $m \times n$ matrix A and the $n \times p$ matrix B results in a $m \times p$ matrix.

Example: If A is a 4×3 matrix and B is a 3×5 matrix, then

- Its product AB will be a 4×5 matrix
- The entry in the 1st row and 1st column will be the product of the 1st row of A and the 1st column of B
- ...
- The entry in the i^{th} row and j^{th} column will be the product of the i^{th} row of A and the j^{th} column of B



Row-Column rule for the (i, j) -entry of a Matrix Product:

To compute the (i, j) -entry of a Matrix Product AB , locate the i^{th} row of A and the j^{th} column of B as in the diagram. Moving across the the i^{th} row of A and the j^{th} column of B , multiply each entry of the row by the corresponding entry of the column. The sum of these products is the (i, j) -entry AB :

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Diagram illustrating the Row-Column rule for the (i, j) -entry of a Matrix Product AB .

Matrix A (size $m \times n$) is shown with Row i highlighted in blue. The entries in Row i are $a_{i1}, a_{i2}, \dots, a_{in}$.

Matrix B (size $n \times p$) is shown with Column j highlighted in blue. The entries in Column j are $b_{1j}, b_{2j}, \dots, b_{nj}$.

The product $(AB)_{ij}$ is calculated as the dot product of Row i of A and Column j of B :

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Handwritten notes indicate the dimensions: $m \times n$ for A and $n \times p$ for B . A large bracket on the right indicates the resulting matrix AB is of size $m \times p$, with the (i, j) -entry highlighted.

Example: Let $A = \begin{bmatrix} -3 & 3 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & x & 1 \\ y & -3 & z \end{bmatrix}$

1. $AB = \begin{bmatrix} -3 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & x & 1 \\ y & -3 & z \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} -3 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ y \end{bmatrix} & \begin{bmatrix} -3 & 3 \end{bmatrix} \begin{bmatrix} x \\ -3 \end{bmatrix} & \begin{bmatrix} -3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ z \end{bmatrix} \\ \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ y \end{bmatrix} & \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ -3 \end{bmatrix} & \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ z \end{bmatrix} \end{bmatrix}$

2. $BA = \begin{bmatrix} -6 + 3y & -3x - 9 & -3 + 3z \\ 4 - y & 2x + 3 & 2 - z \end{bmatrix}$

No
Answer

2×3
 ~~3×2~~

2×2
 ~~2×2~~



Theorem 2

Let A and B be $k \times m$ matrices, C be an $m \times n$ matrix, and P and Q be $n \times p$ matrices. Then:

1. $(AC)P = A(CP)$ —
2. $C(P + Q) = CP + CQ$ —
3. $(A + B)C = AC + BC$ —
4. $s(AC) = (sA)C = A(sC)$ for any scalar s —
5. $I_k A = A = A I_k$ $\uparrow$



Very Important Point I: Matrix Multiplication is not commutative. In other words, $AB \neq BA$ in general

Example: Let $A = \begin{bmatrix} -3 & 3 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

$$1. AB = \begin{bmatrix} -3 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -6 & -6 & -3+3 \\ 4 & 2 & 2-1 \end{bmatrix} = \begin{bmatrix} -12 & 0 \\ 6 & 1 \end{bmatrix}$$

$$2. BA = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -3 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -6 & 2 & 6-1 \\ 6 & -1 & -6-1 \end{bmatrix} = \begin{bmatrix} -4 & 5 \\ 8 & -7 \end{bmatrix}$$



Very Important Point II: Cancellation laws do not hold in general for matrix multiplication

Example: Let $A = \begin{bmatrix} -3 & 3 \\ 2 & -2 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}, C = \begin{bmatrix} 4 & 5 \\ 0 & 5 \end{bmatrix}$

1. $\underline{AC} = \begin{bmatrix} -12 & 0 \\ 8 & 0 \end{bmatrix}$

2. $\underline{BC} = \begin{bmatrix} -12 & 0 \\ 8 & 0 \end{bmatrix}$

$$AC = BC$$
$$A \neq B$$



Very Important Point III: If the product of matrices is the zero matrix, you cannot conclude that either of your original matrices are zero

Example: Let $A = \begin{bmatrix} -3 & 3 \\ 2 & -2 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$

Then $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$AB = 0$$



Definition: If A is an $n \times n$ matrix, then

$$AA = A^2$$

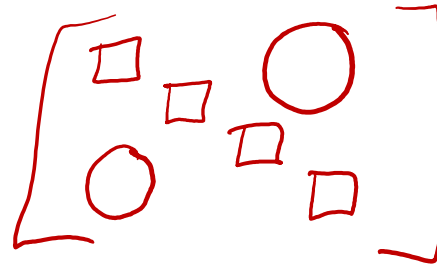
$$AAA = A^3$$

$$AAAA = A^4$$

and so on.

Also, $A^0 = I_n$

Definition: The (i, j) -entry of a matrix A is a **diagonal entry** if $i = j$. The set of diagonal entries is the **diagonal** of A . A square matrix A is a **diagonal matrix** if all non-diagonal entries are 0.



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



The **transpose** of a matrix A , denoted A^T , is found by interchanging the rows and columns of a matrix

If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix

$$A = \begin{bmatrix} 4 & -2 & -2 \\ 5 & -1 & 3 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 4 & 5 \\ -2 & -1 \\ -2 & 3 \end{bmatrix}$$



Theorem 3

Let A, B be $m \times n$ matrices, and let s be any scalar. Then

a) $(A^T)^T = A$ 

b) $(A + B)^T = A^T + B^T$ 

c) $(sA^T)^T = s(A^T)$ 

d) $(AB)^T = B^T A^T$ 



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